

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

A lamp that runs on a glass of water, salt

A Philippines based company has developed a lamp that can run for eight hours at a stretch on a glass of water and two tablespoons of salt. The Sustainable Alternative Lightning (SALT) lamp does not have any hazardous material or component and it has a USB port for charging a smartphone.

According to SALT website, "There are more than 7000 islands in Philippines and most of these do not have access to electricity. We want to eliminate the sustaining cost in areas that rely on kerosene/battery-powered lamps and candles as their main source of lighting."

The lamp uses the science behind the galvanic cell, the basis for battery-making, changing the electrolytes to a non-toxic, saline solution, hence making the entire process safe and harmless.

When electrodes are placed in the electrolyte, the energy generated kicks an LED light into gear. Moreover, the salinity of ocean water can operate the lamp.

Stronger, tougher paper could lead to flexible electronics

Cellulose fibres can be used to make tougher and stronger paper that may pave the way for flexible electronics such as paper smartphones, printable solar cells and green vehicles.

Researchers at University of Maryland, USA, have discovered that paper made of cellulose fibres is tougher and stronger the smaller the fibres get. They explored the mechanical properties of cellulose, the most abundant renewable bio-resource on Earth, and made papers with several sizes of cellulose fibres, ranging in size from 30nm to 10nm.

The paper made of 10-nanometre-thick fibres was 40 times tougher and 130 times stronger than regular notebook paper, which is made of cellulose fibres a thousand times larger.

Bendy liquid metal coils for making stretchable loudspeakers

Coils of liquid metal could be used to make stretchable loudspeakers and microphones, potentially leading to new kinds of hearing aids, heart monitors, and wearable and implantable devices.

Acoustic devices often rely on rigid metal coils that can both emit and detect sound. Scientists in Korea have created a stretchable acoustic device by replacing this rigid coil with a deformable, liquid metal coil. This new coil is made of Galinstan, a highly conductive liquid metal alloy of gallium, indium and tin. They used a syringe to inject Galinstan into a spiral channel in a thin film of flexible silicone rubber. They then attached copper wires to the end of the coil and a neodymium magnet (made from an alloy of neodymium, iron and boron) to the centre of the coil.